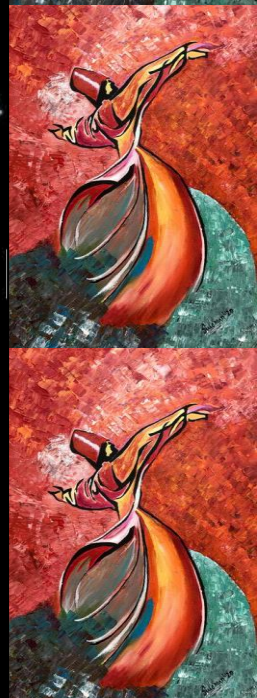




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



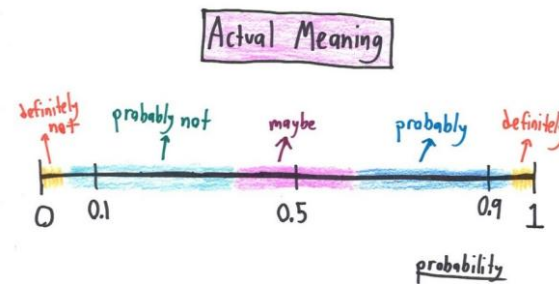


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 5 - Lecture No 09 – February 09, 2026



Announcements!

1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. Quiz No 03 (Unit 3, Counting) – February 11, 2026
3. Midterm Examination – March 3, 2026, from 03:25 PM to 04:55 PM at the Tariq Rafi Lecture Hall
4. Assignment No 01 – due date February 10, 2026 (11:59 PM)
 - Upload a single scanned(clear) PDF on Canvas and also submit a hard copy (Wednesday class) – late submission will not be handled.
 - If you have any questions, feel free to contact Dr Muhammad Sami on MS-Teams.

Agenda for today

1. Unit 3 – Counting (complete)
2. Unit 4 – Discrete Random Variables (start)

TAs -

EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics, Spring 2026



Meet Your TAs Zain Naqi

About Me

I am a Computer Science Junior. My career interests include machine learning, deep learning, and reinforcement learning. Apart from academics, I do competitive programming and run a small e-commerce business. Mostly, you will find me in the library discussion area.

Ehsaas Hours

- * Tue 9:30 - 11:30
- * Thu 9:30 - 11:30

Contact Me

✉ zno9224@st.habib.edu.pk
or "Zain Naqi" on Teams



Meet Your TAs Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

- * Mon 4:00-5:00
- * Wed 4:00-5:00
- * Fri 3:30-4:30

Contact Me

✉ rso8770@st.habib.edu.pk
Or "Rayyan Ali Shah" on Teams



Meet Your TAs Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I'm not in class, I'm most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

- * Mon 2:30 - 4:00
- * Wed 2:30 - 4:00
- * Fri 12:00-1:00

Contact Me

✉ hao9269@st.habib.edu.pk
on Teams or Outlook

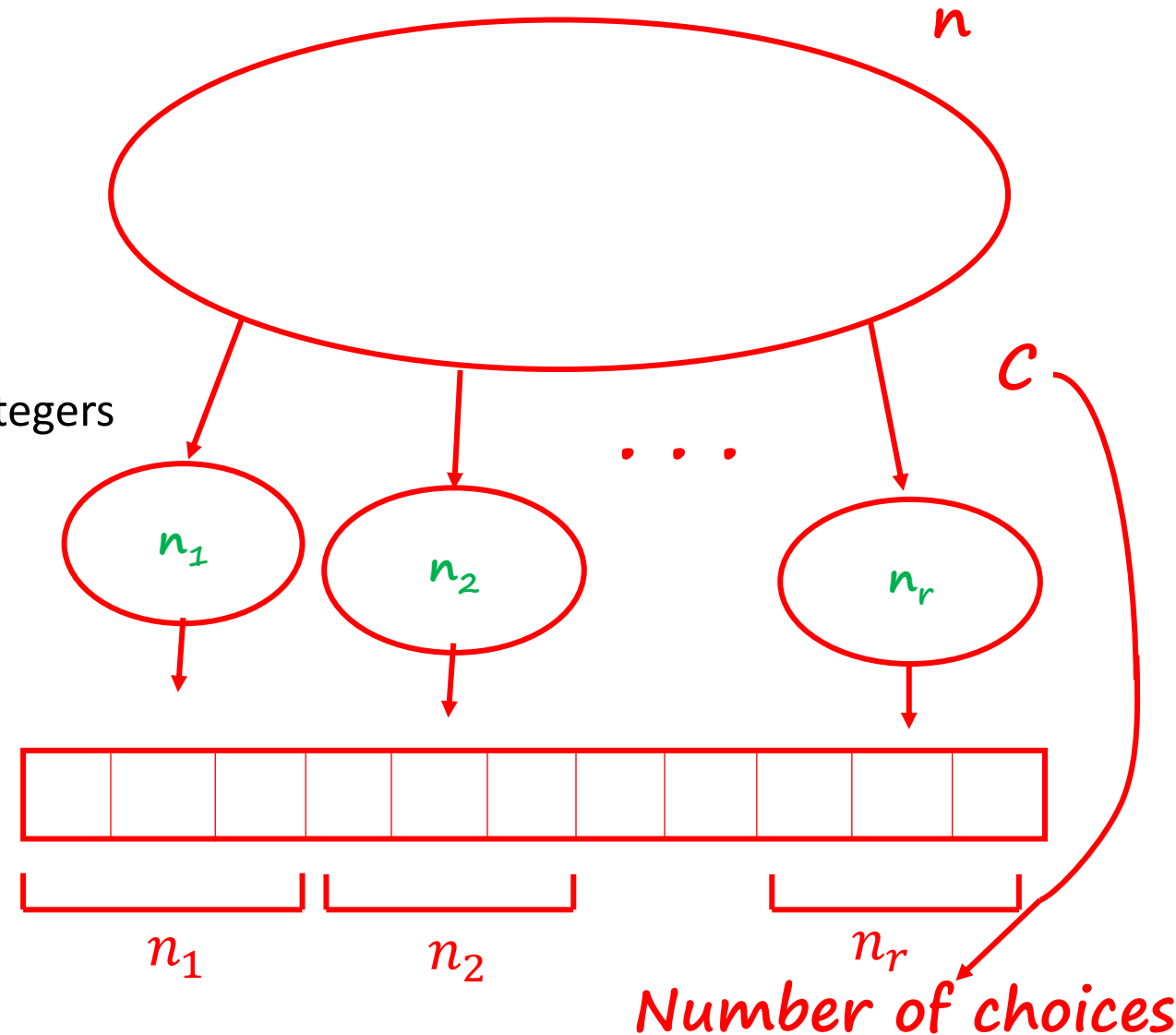
Partitions

- $n \geq 1$ distinct items; $r \geq 1$ persons
Give n_i items to person i
 - Here $n_1, \dots, n_r$ are given nonnegative integers
 - With $n_1 + \dots + n_r = n$ are given nonnegative integers
- Ordering n items: $n!$
 - Deal n_i to each person i , and then order

$$C \cdot n_1! n_2! \dots n_r! = n!$$

Solving for C we get:

$$\text{Number of Choices, } C = \frac{n!}{n_1! n_2! \dots n_r!} \quad (\text{multinomial coefficient})$$



Example: 52-card deck, dealt (fairly) to four players.
Find $P(\text{each player gets an ace})$

➤ Total possible Number of outcomes: $\frac{52!}{13! 13! 13! 13!}$

➤ Constructing an outcome with one ace to each person:
(Distribute the aces) $\frac{4!}{1! 1! 1! 1!} = 4 * 3 * 2 * 1$

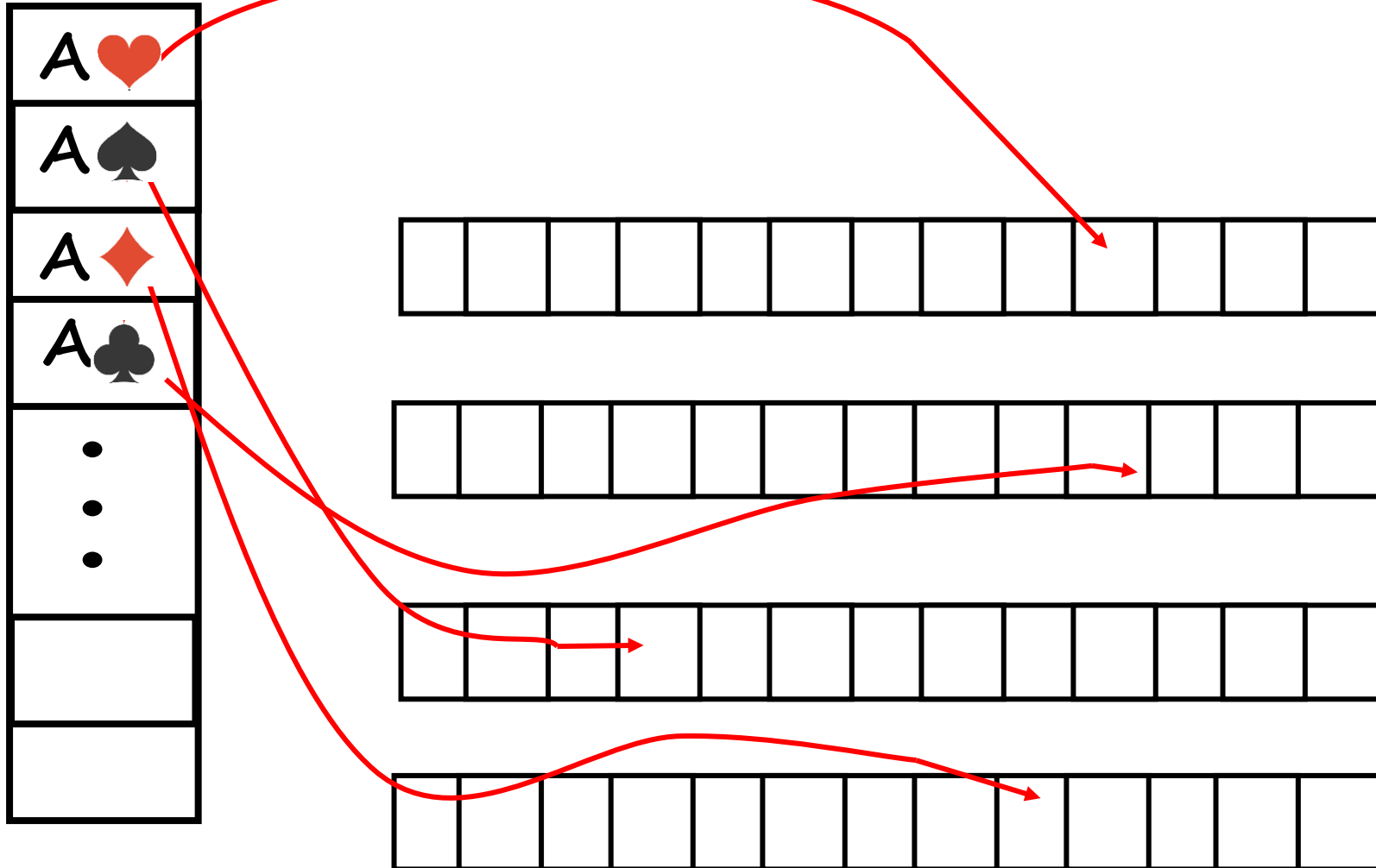
➤ Distribute the remaining 48 cards: $\frac{48!}{12! 12! 12! 12!}$

➤ Answer: $P(\text{each player gets an ace}) = \frac{\frac{4!}{1! 1! 1! 1!} * \frac{48!}{12! 12! 12! 12!}}{\frac{52!}{13! 13! 13! 13!}}$

Example: 52-card deck, dealt (fairly) to four players.
Find $P(\text{each player gets an ace})$

A smart solution

Stack the deck, with 4 aces on top



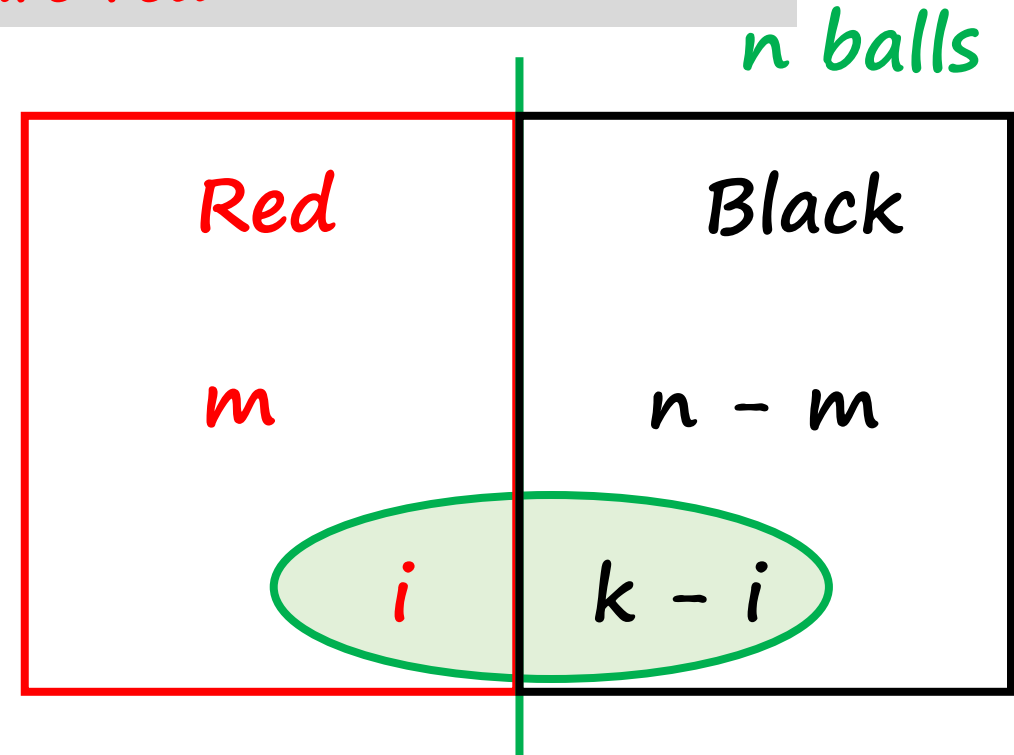
$$\begin{aligned} & A♥ \quad A♠ \quad A♦ \quad A♣ \\ & \frac{52}{52} * \frac{39}{51} * \frac{26}{50} * \frac{13}{49} \\ & = 0.105 \end{aligned}$$

Counting Example

We have a total of n balls, out of which m are red balls. The remaining $(n-m)$ balls are black balls

→ We have to pick k balls such that i balls are red

Find the Probability,
 $P(i \text{ red balls}) = P_r$



Counting Example

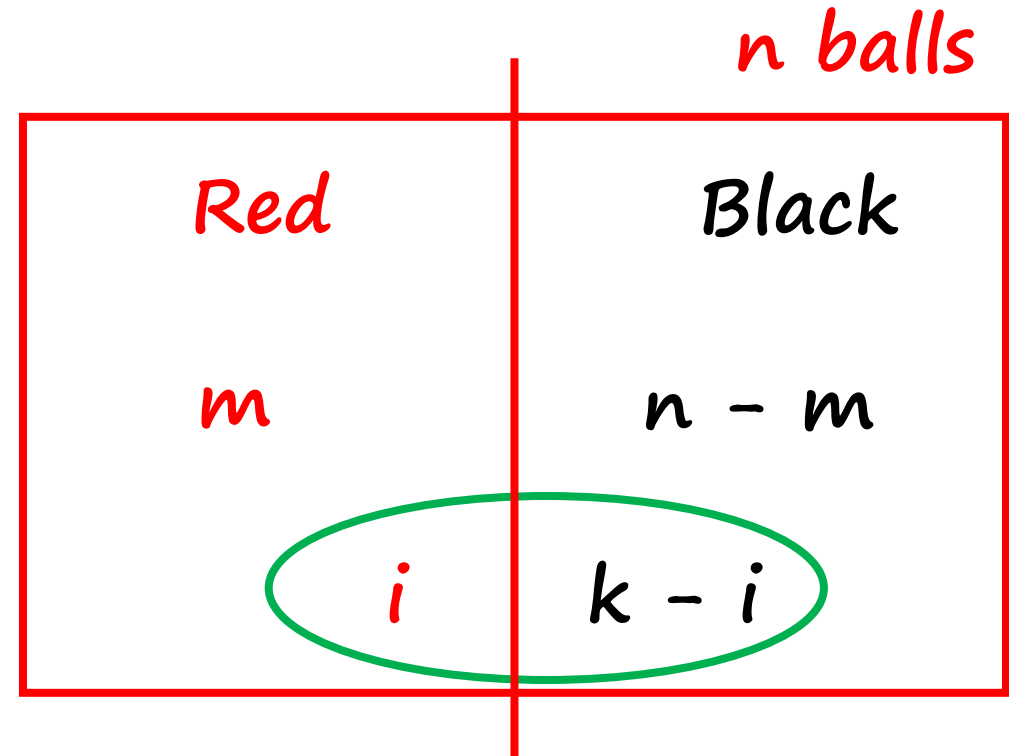
$$P(i \text{ red balls}) = P_r$$

$$|\Omega| = \binom{n}{k}$$

➔ Let $C = \#$ of ways to get i red balls in k balls
 $= a * b$

- $a = \#$ of ways to get i out of m red balls
- $b = \#$ of ways to get $(k - i)$, out of $(n - m)$, black balls

$$P_r = \frac{C}{|\Omega|} = \frac{\binom{m}{i} \binom{n-m}{k-i}}{\binom{n}{k}}$$

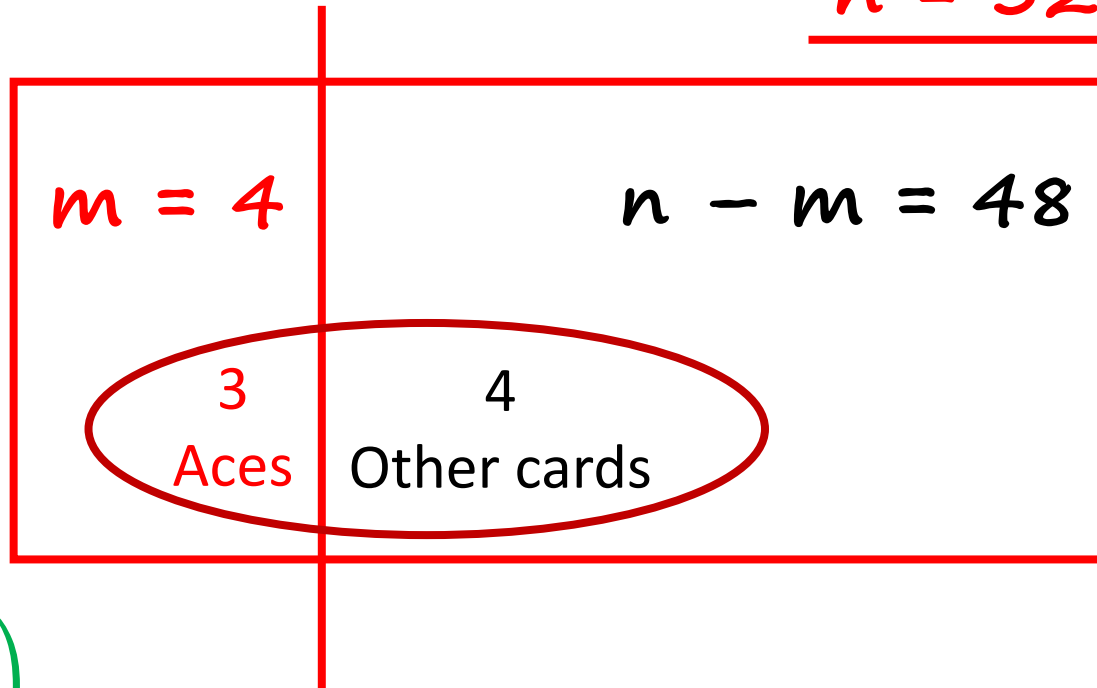


Counting Example

In a deck of 52 cards, you select 7 cards.
What is the probability that out of 7 cards three are aces, $P(3 \text{ aces})$?

$$P_r = \frac{C}{|\Omega|} = \frac{\binom{m}{i} \binom{n-m}{k-i}}{\binom{n}{k}}$$

$$\underline{n = 52}, \quad \underline{k = 7}, \quad \underline{i = 3}$$



$$|\Omega| = \binom{n}{k}$$

$$C = a * b$$

$$P(3 \text{ aces}) = \frac{\binom{4}{3} \binom{48}{4}}{\binom{52}{7}}$$

The multinomial Probabilities

- Balls of different colors: $i = 1, \dots, r$
- Probability of picking a ball color i is p_i
- Draw n balls, independently
- Given nonnegative numbers n_i , with $n_1 + \dots + n_r = n$
- Find: $\mathbf{P}(n_1 \text{ balls of color 1, } n_2 \text{ balls of colors 2, } \dots, n_r \text{ balls of color } r)$
- Special case $r = 2$; colors: “heads”, “tails”

$n_1 \longleftrightarrow k \text{ heads}$

$n_2 \longleftrightarrow n - k \text{ heads}$

$$p_1 = p$$

$$p_2 = 1 - p$$

$$\binom{n}{k} p^k (1 - p)^{n-k}$$

$$\frac{n!}{n_1! n_2!} p_1^{n_1} p_2^{n_2}$$

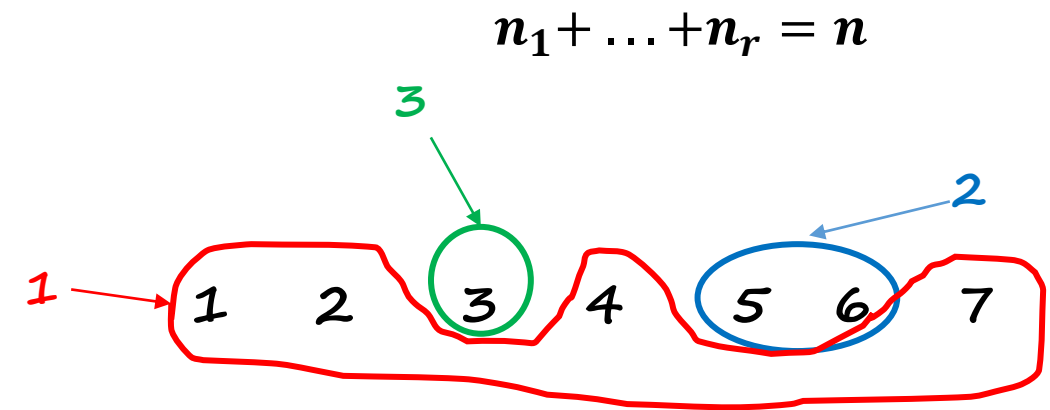
The multinomial Probabilities

$$r = 3$$

$$n = 7$$

1 1 3 1 2 2 1

Type: (4, 2, 1)



$$\text{Prob} = p_1 \cdot p_1 \cdot p_3 \cdot p_1 \cdot p_2 \cdot p_2 \cdot p_1 = p_1^4 \cdot p_2^2 \cdot p_3$$

$$P(\text{particular sequence of "type"}(n_1, n_2, \dots, n_r)) = p_1^{n_1} p_2^{n_2} p_3^{n_3} \dots p_r^{n_r}$$

sequence of "type"($n_1, n_2, \dots, n_r$) $\longleftrightarrow$ Partition of $\{1, \dots, n\}$
into subsets of sizes $n_1, n_2, \dots, n_r$

$$P(\text{get type } (n_1, n_2, \dots, n_r)) = \frac{n!}{n_1! n_2! \dots n_r!} p_1^{n_1} p_2^{n_2} p_3^{n_3} \dots p_r^{n_r}$$

Practise Problem 1 –

- ❖ An experiment consists of rolling a 20 sided die three times. The number on top of each die is recorded. The numbers are written down in the order in which they are observed. How many possible ordered triples of numbers can result from the experiment? (Note the triple (17, 10, 3) is not the same result as the triple (3, 10, 17)).

Answer:–

There are 20 ways each throw can come up and the order is important so the answer is $20 \cdot 20 \cdot 20 = 20^3 = 8000$.

Practise Problem 2

- ❖ When you buy a Powerball ticket, you select 5 different white numbers from among the numbers 1 through 59 (order of selection does not matter), and one red number from among the numbers 1 through 35. How many different Powerball tickets can you buy?

Answer:-

you need to select 5 distinct white numbers, so you can do this $C(59, 5) = 5,006,386$ ways. Then you can pick the red number $C(35, 1) = 35$ ways so the total number of tickets is:-

$$C(59, 5) \cdot C(35, 1) = 5,006,386 \cdot 35 = 175,223,510.$$

Practise Problem 3

❖ If I flip a coin 20 times, I get a sequence of Heads (H) and tails (T).
How many different sequences have at least three heads?

Midterm-1
Fall 2019

Solution: The total number of sequences that are possible here is 2^{20} . We want number of sequences that have at least 3 heads, so we can find out the number of sequences that have no heads, 1 heads, and 2 heads and subtract it from total number of sequences.

Number of sequences with no heads = 1

Number of sequences with 1 heads = $\binom{20}{1}$

Number of sequences with 2 heads = $\binom{20}{2}$

Therefore,

$$\text{Number of sequences with at least 3 heads} = 2^{20} - 1 - \binom{20}{1} - \binom{20}{2}$$

We can also approach it by finding the number of sequence with 3 heads, 4 heads, all the way to 20 heads and summing them up, i.e.

$$\text{Number of sequences with at least 3 heads} = \sum_{k=3}^{20} \binom{20}{k}$$

Practise Problem 4

❖ From 5 seniors, 4 juniors, 2 sophomores and 9 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Answer

There are 5 ways to select a senior, 4 ways to select a junior, 2 ways to select a sophomore and 9 ways to select a freshman. This gives $5 \cdot 4 \cdot 2 \cdot 9 = 360$ ways to select a subset with 4 elements containing one member of each class. When you have done this there are $20 - 4 = 16$ members left and you may choose any one of these to round out the group. Hence the answer is $360 \cdot 16/2 = 2,880$. You must divide by 2 because each set of 5 elements selected by this procedure occurs twice

*Practise Problem 4 –
Some more explanation*

❖ From 5 seniors, 4 juniors, 2 sophomores and 9 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Answer

Here is another approach.

Because there are 5 members in the subset and 4 classes, exactly one class occurs twice.

- If there are 2 seniors, these can be selected in $C(5, 2)$ ways and the set filled out with 1 junior, 1 sophomore and 1 freshman, hence in $C(5, 2) \cdot 4 \cdot 2 \cdot 9 = \underline{720 \text{ ways}}$.

- If there are 2 juniors, these can be selected in $C(4, 2)$ ways and the set filled out with 1 senior, 1 sophomore and 1 freshman, hence in $5 \cdot C(4, 2) \cdot 2 \cdot 9 = \underline{540 \text{ ways}}$.

- If 2 sophomores, $5 \cdot 4 \cdot C(2, 2) \cdot 9 = \underline{180}$.

- If 2 freshmen, $5 \cdot 4 \cdot 2 \cdot C(9, 2) = \underline{1,440}$.

Hence the answer is $720 + 540 + 180 + 1,440 = 2,880$.

Practice Problem No 5: Recall that a standard deck of cards has 52 cards. The cards can be classified according to suits or denominations. There are 4 suits, hearts, diamonds, spades and clubs. There are 13 cards in each suit. There are 13 denominations, Aces, Kings, Queens, ,Twos, with 4 cards in each denomination. A poker hand consists of a sample of size 5 drawn from the deck. Poker problems are often like urn problems, with a hitch or two.

- Part a: How many poker hands consist of 2 Aces and 3 Kings?
- Part b: If a poker hand has three cards from one denomination and two from another it is called a full house. Find the Probability of getting a Full House, P_{FH} .
- Part c: A royal flush is a hand consisting of an Ace, King, Queen, Jack and Ten, where all cards are from the same suit. Find the Probability of getting a royal flush, P_{RF} .

Practice Problem No 5: Recall that a standard deck of cards has 52 cards. The cards can be classified according to suits or denominations. There are 4 suits, hearts, diamonds, spades and clubs. There are 13 cards in each suit. There are 13 denominations, Aces, Kings, Queens, ,Twos, with 4 cards in each denomination. A poker hand consists of a sample of size 5 drawn from the deck. Poker problems are often like urn problems, with a hitch or two.

- Part a: How many poker hands consist of 2 Aces and 3 Kings?

$$\begin{array}{c} \text{Ace's} \end{array} \quad \binom{4}{2} \times \text{3 Kings} \quad \binom{4}{3} \\ = 6 \cdot 4 = 24 \text{ ways.}$$

- Part b: If a poker hand has three cards from one denomination and two from another it is called a full house. Find the Probability of getting a Full House, P_{FH} .

There are 13 ways to Pick the first denomination and One can choose $\binom{4}{3}$ ways of Select 3 Cards of that denomination. There are 12 ways of Select the 2nd denomination. and $\binom{4}{2}$ of Select the 2 cards out of 4.

$$= 13 \cdot \binom{4}{3} \times 12 \times \binom{4}{2} = 3744$$

$$P_{FH} = \frac{3744}{\binom{52}{5}} = \frac{3744 \times 5!}{48 \times 49 \times 50 \times 51 \times 52} = 0.00144$$

- Part c: A royal flush is a hand consisting of an Ace, King, Queen, Jack and Ten, where all cards are from the same suit. Find the Probability of getting a royal flush, P_{RF} .

There are 4 Royal flush's in the (3) deck and within each 1 way of selecting the cards.

$$\frac{4}{\binom{52}{5}} = \frac{4 \times 120}{48 \times 49 \times 50 \times 51 \times 52}$$

$$= 1.53 \times 10^{-6}$$